Planning Stage

1. **Project description**
   Our project is a skincare application called Glowly. Many people struggle with various skin problems, and our application aims to help them find suitable solutions. To use the app, users simply upload photos of their skin issues and specify their concerns. The application then analyzes the condition and provides medically based recommendations and guidance on how to treat and improve the skin.

2. **Problem Statement**
   The problem our application aims to solve is that many people sometimes use skincare products that are not suitable for their skin type, which can cause damage instead of improvement. In other cases, users may choose products that are individually effective and suitable, but when used together, they may not work well or could negatively affect the skin. Glowly helps users avoid these issues by analyzing their skin condition and providing personalized recommendations for compatible and appropriate skincare products.

3. **In Scope.**

- Photo Analysis: The app scans the skin photo to identify the problem.
- Weekly Check-up: Every week, the user uploads a new photo. The app tells them whether their skin is getting better or not.
- Product Helper: The user can check a specific product to see if it is safe for their skin.
- Simple Routine: The app provides a daily schedule (morning and night) to improve the condition.

4. **Out of Scope**

- Selling Products: The app will not sell any creams or medicines.
- Doctor Chat: There is no live video call with a real doctor.
- Severe Cases: The app does not treat severe burns or deep wounds.

Functional Requirements

1. User Registration

1.1The user shall be able to create a new account by entering name, email, phone number, and password.
1.2 The system shall verify that the email is valid and not already registered in the database. If the email is already used, the system shall display the message: “Email already exists.”
1.3 The system shall send a verification code to the user’s email or phone number.
1.4 The user shall be able to enter the verification code.
If the code is incorrect, the system shall display the message: “Invalid verification code.” If the code is correct, the system shall create the account successfully.

2. User Login

2.1 The user shall be able to log in using email and password.
2.2 The system shall verify the login credentials.
If the credentials are incorrect, the system shall display the message: “Invalid login credentials.”

3. Skin Information Input

3.1 The user shall be able to enter their skin type (oily, dry, combination, sensitive).
3.2 The user shall be able to specify their skin concerns (e.g., acne, pigmentation, dryness).
3.3 The system shall allow the user to provide information about allergies to specific ingredients or products.
3.4 The user shall be able to upload images of their skin condition.
3.5 The system shall store user information and images in the user profile.

4. Condition Analysis and Product Recommendation

4.1 The system shall analyze the user’s data and upload images to determine skin conditions.
4.2 The system shall recommend medically approved skincare products based on the analysis.
4.3 The system shall display product descriptions and usage instructions.

5. Trial Period

5.1 The system shall assign a trial period for each product ranging from one to two weeks.
5.2 The system shall request the user to evaluate their skin condition after the trial period.
5.3 If irritation or negative effects occur, the system shall recommend alternative products.

6. Progress Tracking
6.1 The user shall be able to upload before and after images of their skin.
6.2 The system shall store user images and history for up to 3 months.
6.3 The system shall display comparisons between before and after images.

7. Routine Update
7.1 The system shall allow updating the skincare routine based on user progress.
7.2 The system shall allow adding or removing products according to skin condition changes.

8. Notifications and Reminders
8.1 The system shall send notifications to remind users to follow their skincare routine.
8.2 The system shall notify users when the trial period ends.

9. Alternative Solutions
9.1 The system shall suggest alternative solutions if side effects or issues occur.
9.2 The system shall allow users to report problems or feedback regarding products.

10. Product Reviews and Feedback
10.1 The user shall be able to rate the products they have used.
10.2 The user shall be able to write feedback describing product effectiveness.
10.3 The system shall display ratings and reviews from other users.
10.4The system shall allow users to view reviews before using any product.

Non-Functional Requirements

1. **Performance**
   The system shall analyze and return skincare results within 5 seconds after image upload under normal network conditions.
   The system shall support at least 1,000 concurrent users without performance degradation.

2. **Security**
   The system shall encrypt all user data (including images) using HTTPS during transmission.
   The system should store user data securely and restrict access to authenticated users only.

3. **Usability**
   The system shall allow new users to upload an image and receive results within 2 minutes without prior training.
   The system shall present results in a clear and understandable format using non-technical language.

4. **Reliability**
   The system shall maintain at least 99% system availability per month.
   The system shall automatically save user data to prevent data loss in case of failure.

5. **Scalability**
   The system shall support scalability through cloud-based infrastructure.
   The system shall allow future feature integration without major system redesign.

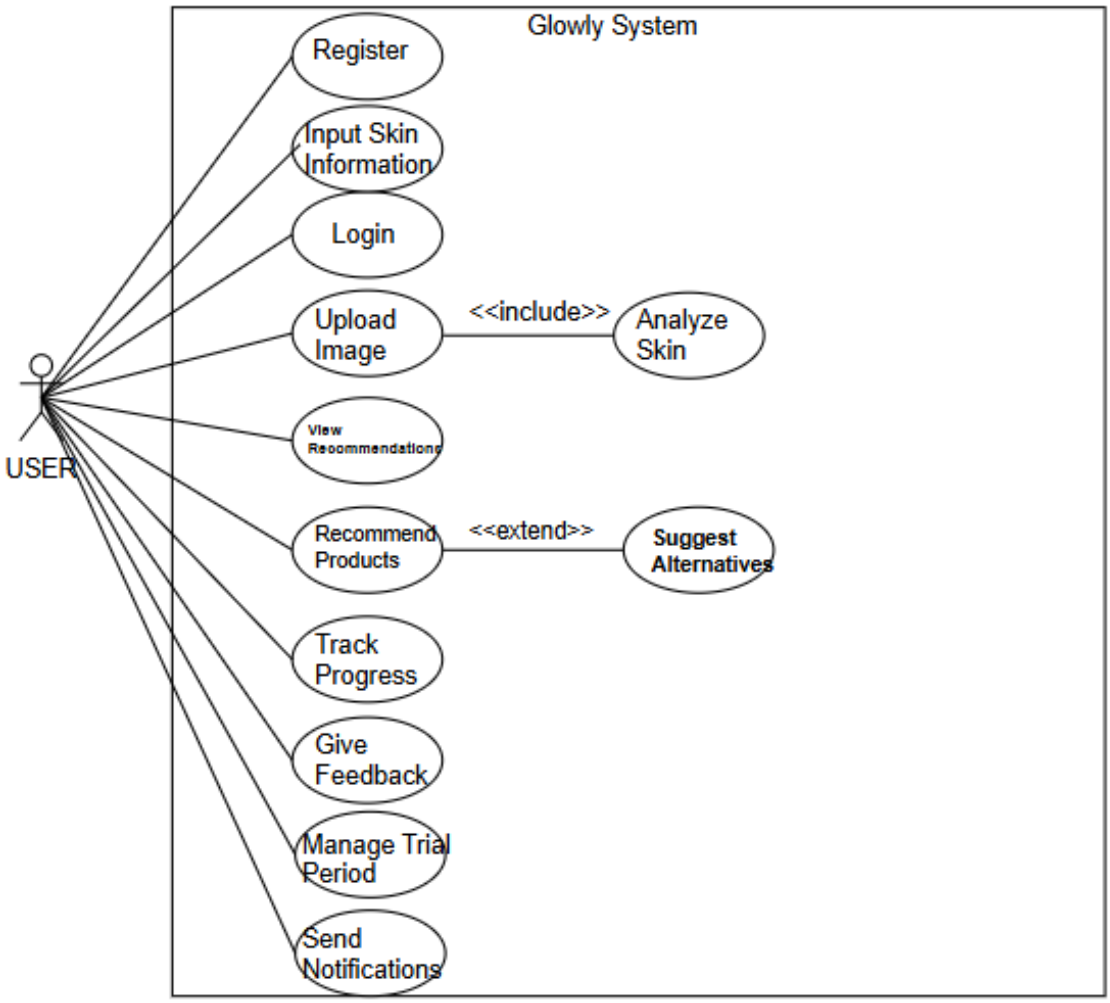
6. **Accuracy**
   The system should provide skin analysis results with at least 85% accuracy.

1. User story

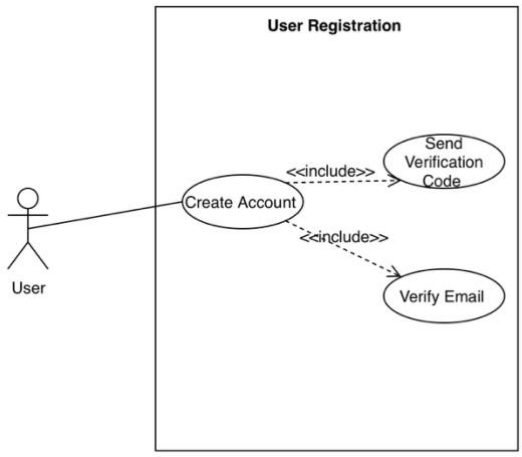
Scenario Name: Analysis and product recommendation

Item	Description
Starting Situation	Skin Analysis and Product Recommendation
Normal Flow	User is logged into the system and wants to get personalized skincare recommendations.
What Can Go Wrong	1. User enters their skin type (oily, dry, combination, sensitive) 2. Users select their skin concerns (acne, pigmentation, dryness, etc.) 3. User provides information about allergies 4. User uploads images of their skin condition 5. System stores the data in the user profile 6. System analyzes the provided data 7. System generates suitable skincare recommendations 8. System displays recommended products with descriptions and usage instructions
Concurrent Activities	• User enters incomplete information • Uploaded image is unclear or unsupported format • System fails to analyze the data • No suitable products found
Final State	Users receive personalized skincare product recommendations based on their skin condition

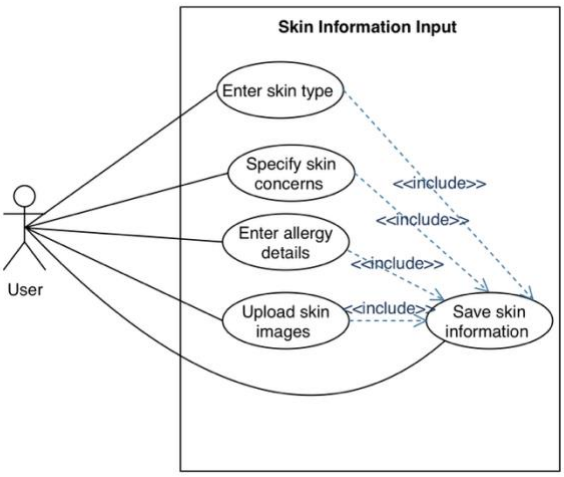
2. Use Case Diagram (Full System)



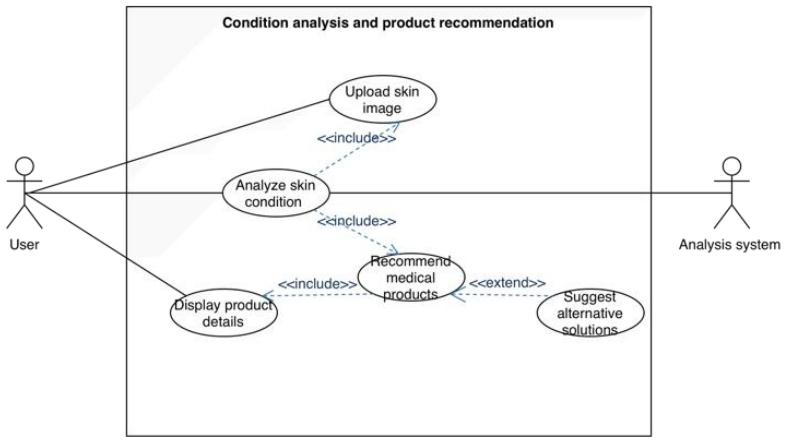
3. Use Case Description (Table Format)



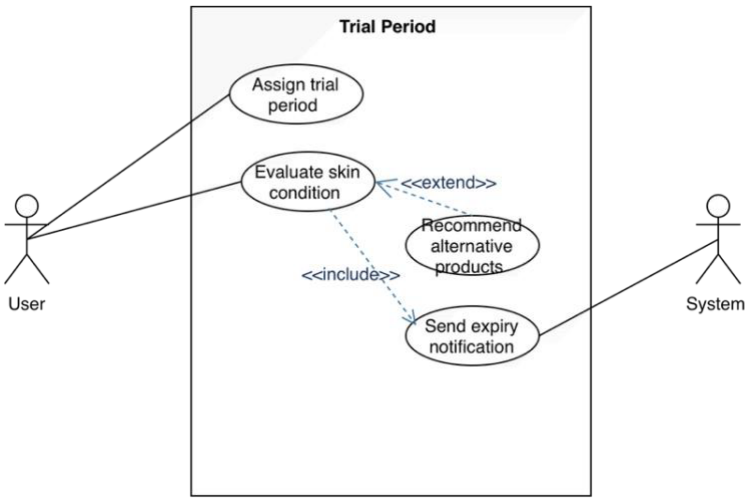
Use Case Name	Actor(s)	Description	data	Stimulus	Response	Comments
User Registration	User	Allows new users to create an account to access the system.	Name, Email, Phone, Password, V-Code.	User clicks the 'Sign Up' button.	System sends a code and creates the account.	Verification is required to ensure email validity.



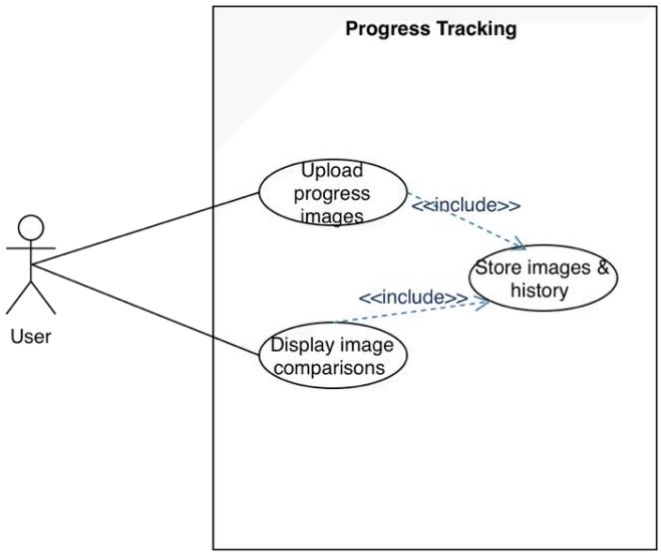
Use Case Name	Actor(s)	Description	data	Stimulus	Response	Comments
Skin Information Input	User	Users provide their skin details and upload images for analysis.	Skin type, concerns (e.g., acne), allergies, and images.	User selects 'Fill Skin Profile' or 'Upload Photo'.	System stores the data and images in the user profile.	Accurate data leads to better medical recommendations.



Use Case Name	Actor(s)	Description	data	Stimulus	Response	Comments
Condition Analysis & Recommendation	User, System	System analyzes skin data to recommend medical products.	Analysis algorithms, product database, user skin info	User clicks the 'Analyze' or 'Get Results' button.	Displays recommended products and usage instructions.	Recommendations are medically approved.



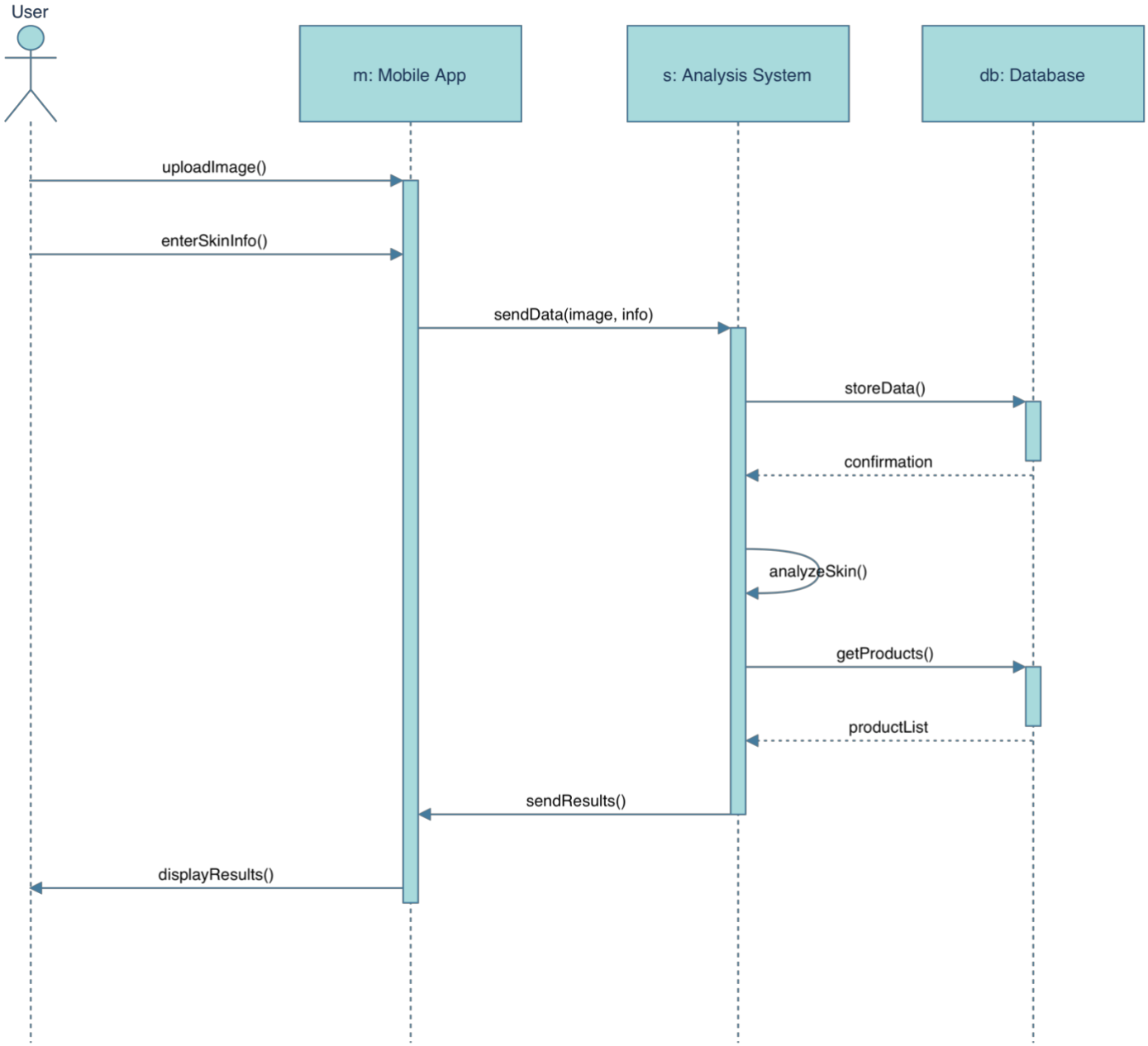
Use Case Name	Actor(s)	Description	data	Stimulus	Response	Comments
Trial Period	System, User	System assigns a period for testing products and evaluates results.	Trial duration (1-2 weeks), user evaluation feedback.	System triggers a request for evaluation after the period.	Suggests alternative solutions if irritation occurs.	Helps identify product sensitivity early.



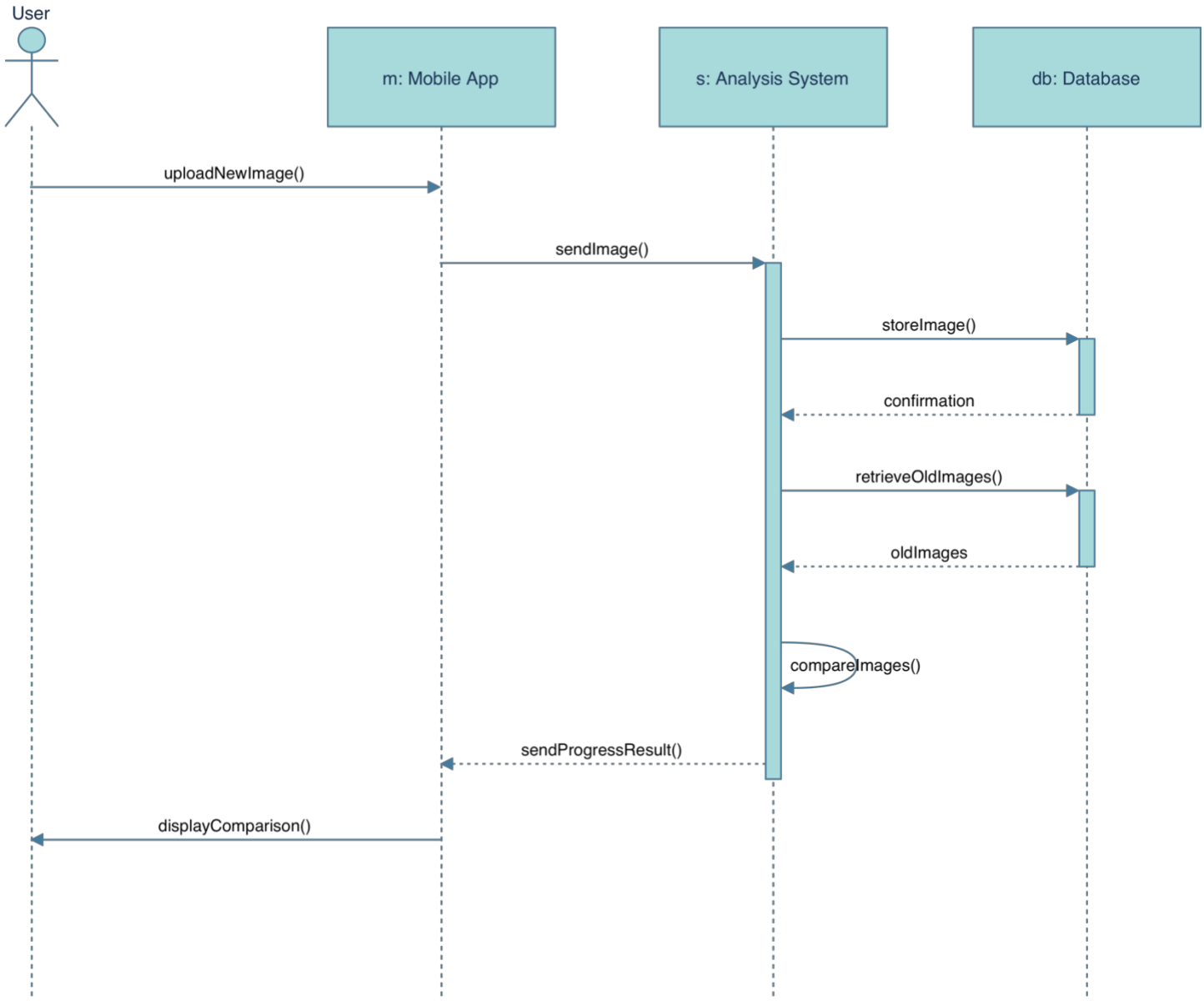
Use Case Name	Actor(s)	Description	data	Stimulus	Response	Comments
Progress Tracking	User	Tracks skin improvement through before/after image comparisons.	Current and historical skin images (up to 3 months).	User clicks 'Compare Progress' or 'View History'.	Displays side-by-side images and a progress report.	Stores user history for a maximum of 3 months.

4. Sequence Diagram (Two Use Cases)

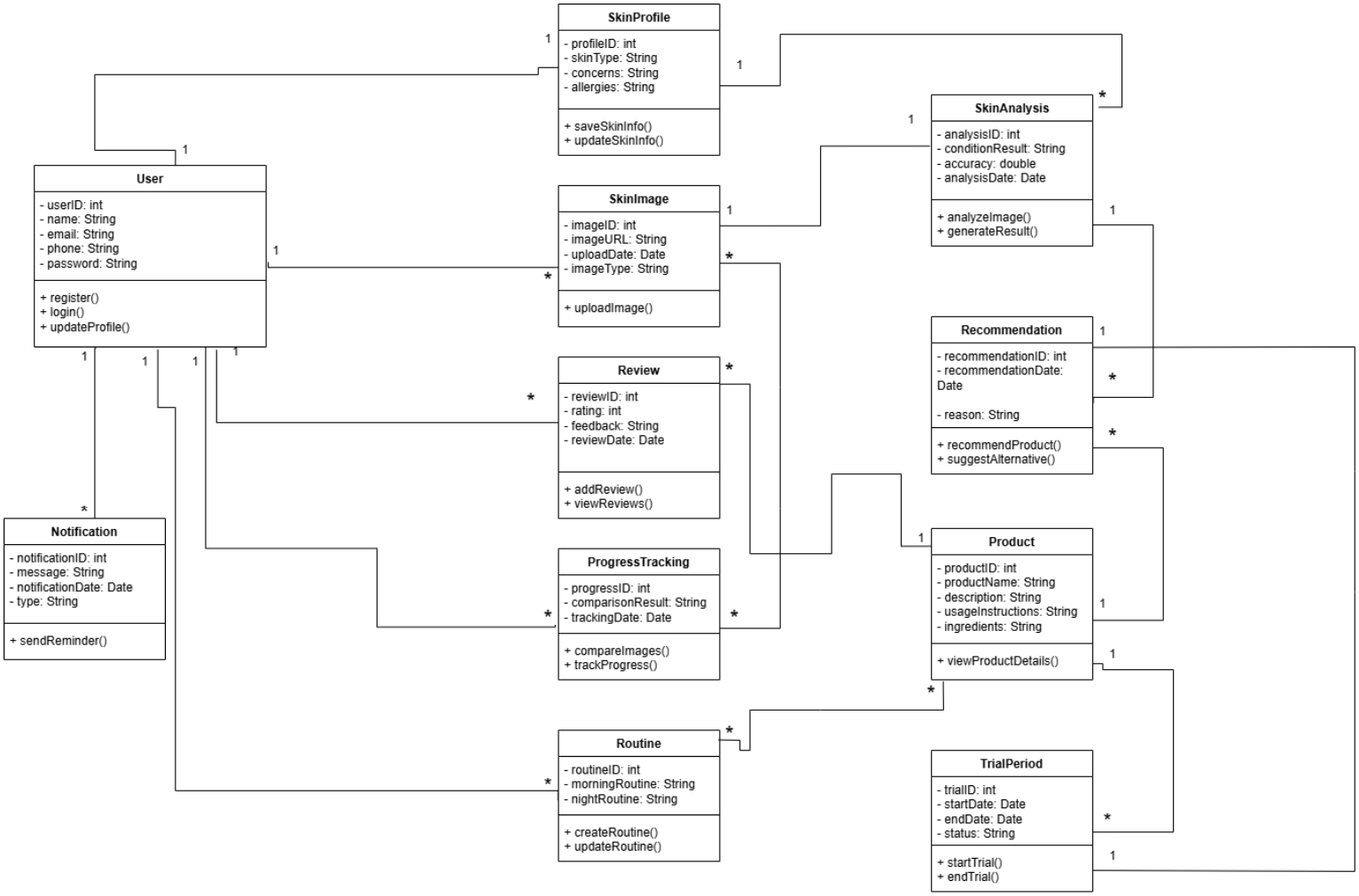
1. Analyze Skin



2. Track Progress



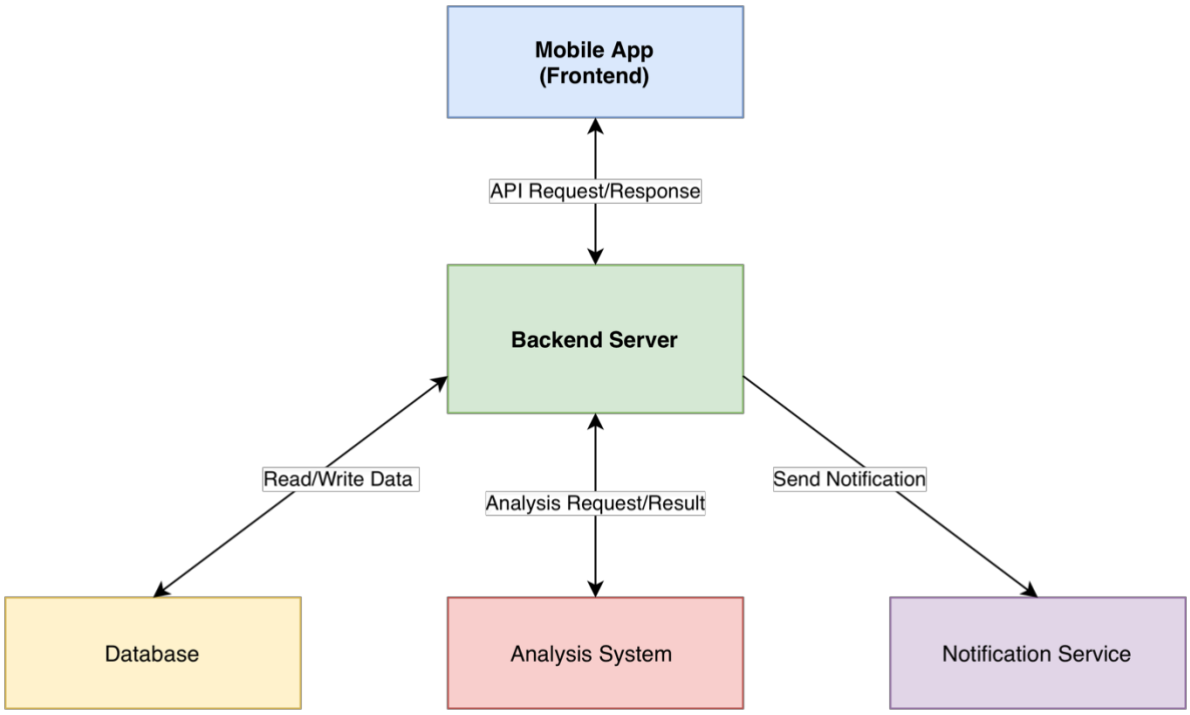
5. Class Diagram



Architectural Pattern: Client-Server

The Glowly system follows a client-server architecture where the mobile application acts as the client and communicates with the server. The server is responsible for processing user data, analyzing skin images, and generating recommendations. This pattern is suitable because it allows centralized data processing, supports multiple users, and ensures better scalability and security.

System Architecture Diagram



The system consists of a mobile application (frontend), a backend server, a database, an analysis system , and a notification service. The mobile app sends requests to the backend server, which processes data, communicates with the database and analysis module, and returns results. The notification service is used to send alerts to users.